NAME: S. MAGALAKSHMI

REG No. 192524144

SUBJECT: Cloud Computing

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SHORT ANSWERS

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing
2. Trace the evolution of cloud computing from hardware evolution to internet Software evolution, and Show how these changes enabled modern cloud platforms
3. Explain the role of Server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a Service model.
4. Discuss the role of ~~Saa~~ data centers in delivering cloud Services.
5. Differentiate IaaS, PaaS, SaaS and XaaS with one Suitable real-world example for each model.

1. Cloud Computing

Cloud computing Service Such as Servers, Storage, databases, networking and Software over the internet. Instead of purchasing and maintaining physical computers and Servers, users can access computing resources from cloud Service providers whenever required.

Characteristics Of Cloud Computing.

On demand Self-Service

Users can access computing resources whenever needed without contacting the Service provider.

Broad network acuss.

Cloud Servic can be acuss through the internet using Computers, laptops and mobile phones.

Resource Pooling

Cloud provides Share Computing resources among multiple users while keeping their data Separated.

Rapid elasticity:

Resources can be increased or decreased according to demand.

Measured Servic:

Users generally pay according to the Consume.

Difference between Cloud Computing and Traditional on Privise Computing:

Traditional on-premise Computing requires organization to purchase, install, maintain and upgrade their own hardware, Cloud Computing allows organization to acuss resources ove the internet without having to own all the underlying Infrastructure.

Conclusion

Cloud Computing is flexible, Scalable, accessible and cost-effective.

Mainframe Computing

Early computers were large and expensive. Multiple users accessed a central computer through terminal

Client - Server Computing

Computers communicated with central servers to share data, application and resources

Server Evolution

Internet Evolution

The growth of the internet and the World Wide Web allowed users to access applications and information from anywhere

Virtualization

Server virtualization allowed one Physical Server to run multiple virtual machines.

This improved hardware utilization and reduced the cost of maintaining physical servers.

Distributed and Data Center Technologies.

Large data centers started combining thousands of servers, storage systems, and networking equipment. These resources could be managed centrally and provided to many users.

Internet - Based Software.

Software began to be delivered through the Internet instead of being installed locally. This lead to SaaS

Mordern Cloud Computing

The Combination of high-speed internet, virtualization, data centers, distributed computing, automation and Internet based Software enabled modern cloud platforms.

Conclusion:

The evolution of hardware and Internet technologies made Computing resources more Scalable, accessible, flexible, and cost-effective, which ultimately enabled modern cloud platforms.

3.

Server Virtualization

It is a technology that divides one physical Server into multiple virtual machines using Software called a hypervisor.

Each Virtual machine can have its own operating System, applications, memory and Storage.

Role Of Server Virtualization in Cloud Computing

1. Better resource Utilization.

   Multiple virtual machines can run on one physical Server, reducing unused hardware capacity.

2. Cost reduction

   Organizations need fewer physical Servers, reducing hardware, electricity, cooling & maintenance costs

3. Scalability

   Virtual machines can be created, modified or removed according to requirements.

Virtualization	Cloud Computing
It is a technology	It is a servic delivery model
Creates virtual resources from physical resources	Delivers computing resources through a network.
Mainly focuses on efficient resource utilization	Focuses on providing resources as services.
Example: VMware,	Example: Google Cloud.

Conclusion.

It is one of the major technologies that enables Cloud Computing because it provides flexibility, scalability & efficient hardware utilization

4. **Roles Of Data Centers in Cloud Computing.**

1. Computing resources.

Data centers contain powerful servers that provide processing power for cloud applications and services.

2. Data Storage

They provide large-scale storage system for storing files, databases, applications and other data.

3. Networking

Networking equipment ~~contact~~ connects servers and storage systems and provide connectivity between the data center and users through the Internet.

Scalability

Data centers contain large pools of resources. Cloud providers can allocate additional servers and storage when demand increases.

Security

Data ~~ber~~ centers use physical security, access control, monitoring, fire protection and other security mechanisms to protect infrastructure and data.

Simple Working

User → Internet → Cloud Provider → Data Center → Servers/Storage → Response to User

Conclusion:

Data centers are the physical foundation of cloud computing. They provide hardware, storage, networking, security, power & cooling required to deliver cloud services reliably to users.

5)

1. IaaS - Infrastructure as a Service

IaaS provides basic computing infrastructure such as virtual machines, storage, networking and servers through the internet.

The users manages the operating system, applications and data.

PaaS - Platform as a Service

PaaS provides a ready-to-use development and deployment platform. Developers can create and run applications without managing the underlying servers and operating systems.

3. SaaS - Software as a Service

SaaS provides complete software applications through the Internet. Users generally do not need to install or maintain the software.

4. XaaS - Anything as a Service.

XaaS means anything as a service. It is broad concept where different IT resources and services are provided over the Internet on demand.

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